Sri Harsha Reddy Sagili – Projects

GitHub profile - https://github.com/23f2005056

Projects Worked on:

- **Clothing Assistant (Group Project) (RAG CHARTBOT)**

  Clothing Assistant Chatbot with Retrieval-Augmented Generation (RAG) is an AI fashion retail assistant that uses Large Language Models (LLMs) combined with semantic retrieval and recommendation systems to provide customer support and personalized outfit recommendations. The system consists of Policy RAG for store-policy queries and Product RAG for fashion recommendations based on customer preferences and purchase history. It was built using BAAI/bge-base-en-v1.5 embeddings with FAISS and Chroma-DB and deployed with Flask, Stream-lit and Google Cloud Platform (GCP).

  URL - https://github.crom/maddy-git/group-5-DS-AI-Lab-Project.git

- **SEEK Portal (Group Project) (RAG CHARTBOT)**

  The SEEK Portal is an AI-powered learning assistant that uses Retrieval-Augmented Generation (RAG) and intelligent tutoring agents to deliver personalized education. The system integrates Fast API, PostgreSQL, Pinecone, and the APIs of OpenAI to support study planning, assignment assessment, programming mentoring, multilingual content summarization, and contextual conversation. There is also a support assistant that looks at bugs reported by users and identifies possible parts of the project code base that are likely to be error prone. It was built on an agent-based architecture and deployed as a scalable system to provide educational support to students, faculty and administrators.

  URL - https://github.com/vaibhavgupta-iitm/soft-engg-project-jan-2025-se-Jan-36.git

- **IE&SCP (WEB APPLICATION)**

  Influencer Engagement & Sponsorship Coordination Platform is a full-stack web application, developed using Flask, Vue.js, SQLite, Redis, and Celery, designed to facilitate collaborations between sponsors and influencers. The platform employs RESTful APIs and role-based access systems for campaign creation, influencer discovery and ad request management, proof verification and administration controls. It also incorporates asynchronous background jobs, caching and reporting features for scalability and efficiency. The project demonstrates practical implementation of web development, database management, and distributed tasks processing in a collaborative digital marketing environment.

  URL - https://github.com/23f2005056/APP--DEV--2.git

- **Bank Marketing Campaign Analysis (MACHINE LEARNING)**

  The project of Bank Marketing Campaign Analysis aims to predict the success of Bank telemarketing campaigns using machine learning techniques. The project uses data preprocessing, feature engineering, PCA and feature selection to efficiently deal with customer and campaign data. To boost the prediction accuracy, we use a stacking ensemble combining several machine learning models such as Random Forest, Gradient Boosting, SVM and XGBoost. The system has been validated with cross validation and F1 score optimization and it has performed well in identifying potential customers who are likely to subscribe to bank term deposits.

  URL- https://github.com/23f2005056/Predict-the-Success-of-Bank-telemarketing.git

- **Image Reconstruction (Generative AI)**

  The Image Reconstruction Under Corruption project focused on the reconstruction of clean images from heavily corrupted inputs using deep learning based image restoration techniques. The system was trained on pairs of corrupted and clean images so that the model can learn degradation patterns such as noise, blur, masking, geometric distortions, and contrast variations. The prediction accuracy was improved by image preprocessing, data augmentation, hyperparameter tuning, and reconstruction model optimization. The corrupted images were fed through the trained neural network which learned how to restore the missing visual details and recover the original image structure, producing the reconstructed images. This solution won the 1st place in the Kaggle competition.

  URL - https://github.com/23f2005056/Image-Reconstruction.git